

Multimodal Sarcasm Detection



CSCI-535
Project Final Presentation
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The Problem

- Sarcasm can be tricky to detect even for humans
- Mehrabian's Rule: When verbal and nonverbal cues conflict, communication is 55% body language, 38% tone, and only 7% words.
- Single-modality models (text-only) often misinterpret sarcasm.
- Multimodal approaches (text, audio, visual) capture deeper context and nonverbal cues.

Utterance

1) Chandler :

Oh my god! You almost gave me a heart attack!

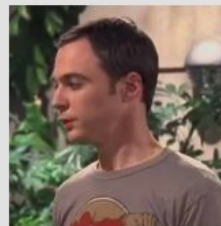
- **Text** : suggests fear or anger.
- **Audio** : animated tone
- **Video** : smirk, no sign of anxiety



2) Sheldon :

Its just a *privilege* to watch your mind at work.

- **Text** : suggests a compliment.
- **Audio** : neutral tone.
- **Video** : straight face.



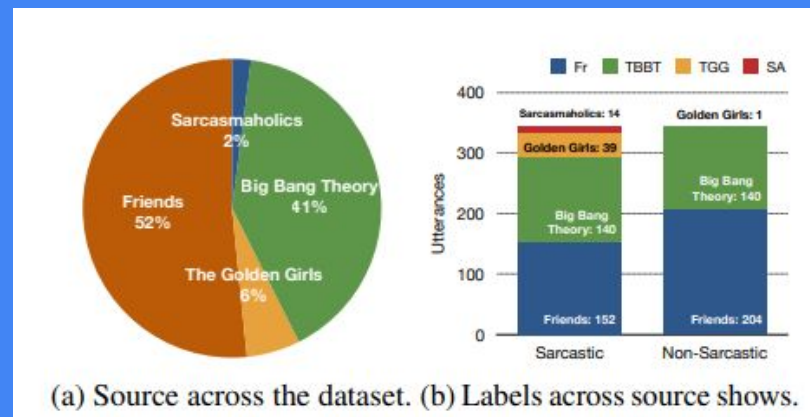
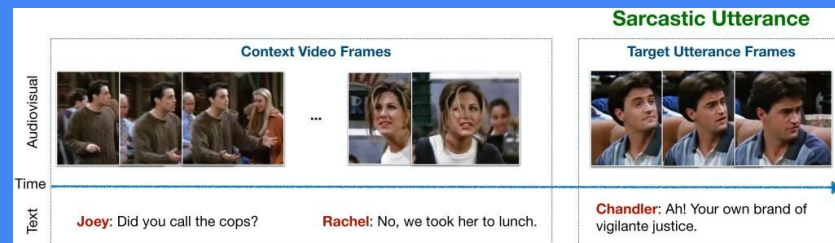
Background

- **Early Text-Based Approaches**
 - **González-Ibáñez et al. (2011)**: Identified lexical cues (hyperbole, sentiment shift) in tweets.
 - **Bamman & Smith (2015)**: Showed context (author, audience) improves sarcasm detection.
- **Audio and Prosodic Features**
 - **Tepperman et al. (2006)**: Early work recognizing sarcasm in spoken dialogue by analyzing pitch, speaking rate, highlighting role of vocal prosody.
 - **Bryant (2010)**: Explored acoustic features like pitch contours and spectral analysis for detecting irony and sarcasm.
 - **Chauhan et al. (2020)**: Showed that combining sentiment, emotion, and sarcasm tasks can improve overall detection accuracy.
- **Visual + Multimodal Advances**
 - **Schifanella et al. (2016)**: Showed images aid sarcasm understanding on social media.
 - **Cai et al. (2019)**: Hierarchical fusion of text, image, and metadata.
 - **Pan et al. (2020)**: Inter-modality attention to bridge text-vision semantic gaps.
- **Multimodal Datasets and Fusion Approaches**
 - **Castro et al. (2019) – MUSTARD Dataset**: Introduced the MUSTARD Dataset from TV shows, for integrated text, audio, and visual cues. Showed that multimodal information performs better than singular modalities
 - **Chauhan et al. (2020)**: Extended MUSTARD with sentiment and emotion annotations, and proposed attention mechanisms to learn relationships across modalities.

Data

MUStARD (Multimodal Sarcasm Detection Dataset) (690 Videos)

- **Source:**
 - TV shows like Friends, Big Bang Theory, etc.
- **Modalities:**
 - Text: Utterance transcriptions with dialogue context
 - Audio: Speech features (tone, pitch, prosody)
 - Visual: Video frames capturing facial expressions and gestures
- **Annotations:**
 - JSON format with keys: utterance, speaker, context, sarcasm (binary label)
- **Purpose:**
 - Enhance sarcasm detection by integrating multiple cues
 - Outperforms text-only methods



Methods

Getting the Modality-wise Embeddings:

1. Audio:

- Used the WavLM (wav2vec2) model for deep feature embedding, Librosa library to extract spectral features such as MFCC, Parselmouth to extract pitch and intensity features, size 1434 feature for non framewise representation
- Framewise slid a 240 ms window over audio at 10 fps, resulting in $772 \times (\text{ctx} \times \text{uttr})$ size embedding per video

2. Video:

- BlazeFace (Google MediaPipe) - 478 3D Facial landmarks/frame & stack together per video
- Non-Framewise variation -
 - i. Uses 1st PCA component to capture max temporal variation across frames
 - ii. Reason- More expressive than simple mean/max pooling
- Tried LibreFace, issues: (extra 12 AU features would have been great)
 - i. Slow mass inference (~16 hrs on CARC A40's); OOMs when trying parallel logic
 - ii. Unstable outputs: errors in landmark & AU extraction across runs for multiple videos -> *huge data loss*

3. Text:

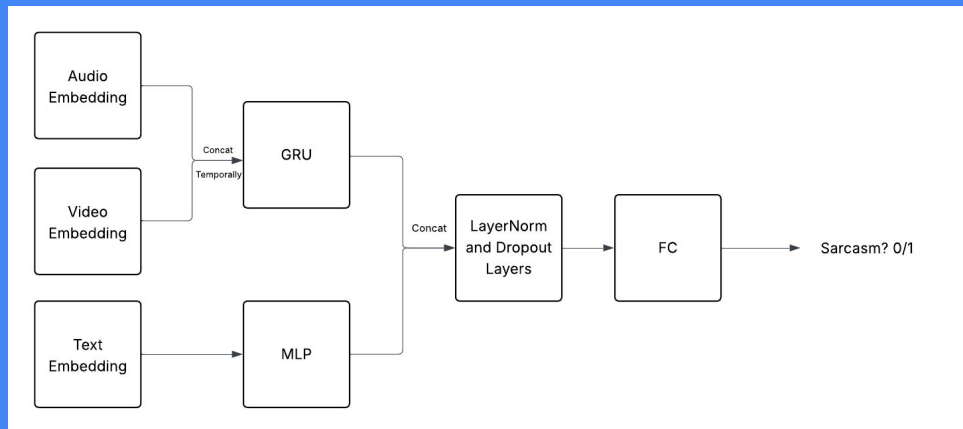
- BERT-base model
 - i. Extracted embeddings from the [CLS] token by averaging its representations from the last four transformer layers.
- Granite-Embedding-278M-Multilingual
 - i. 768-dimensional embedding per context and utterance

● Context handling methods:

- **Text:** Concatenating context and final utterance clips with a separation token
- **Audio-Video:** Concatenating context and final utterance clip & treating them as a single clip

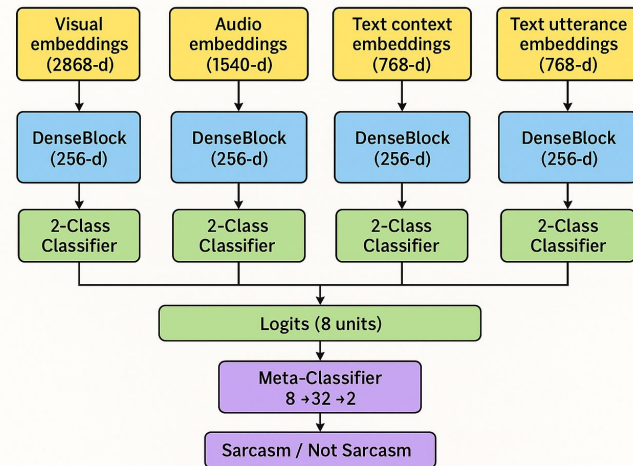
Early Fusion Methods

1. Early Fusion:
 - a. Non-Framewise:
 - MLP-based Neural Network (combo of Linear + ReLU + BatchNorm + Dropout + FC)
 - b. Framewise (in-diagram):
 - Bi-GRU for Aligned Temporal features + MLP for text projection



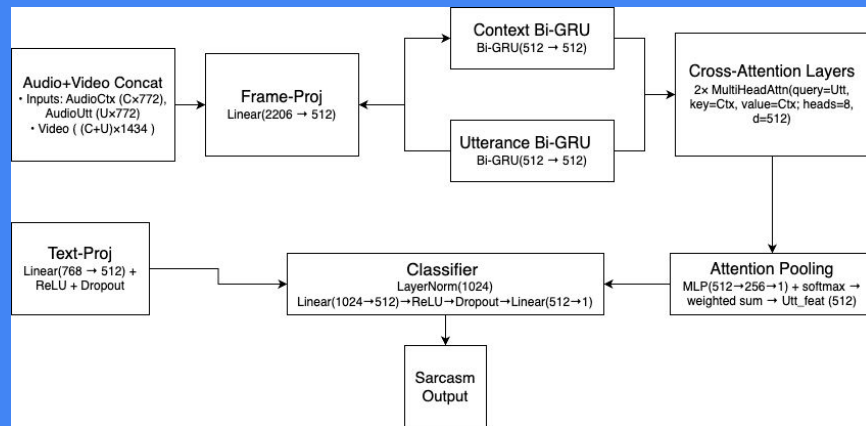
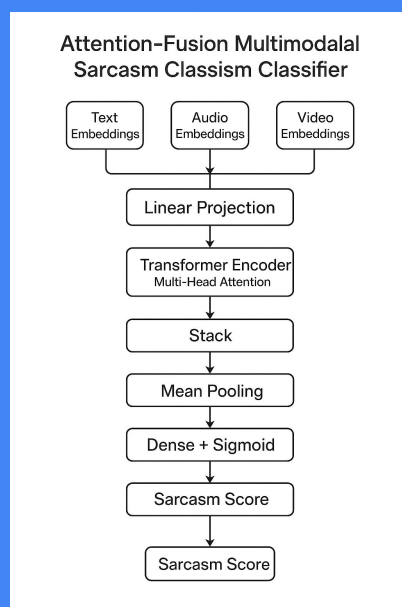
Late Fusion Based Methods

1. Late Fusion:
 - a. Non-Framewise
 - Per-modality MLP encoders $\rightarrow$ 2-logit heads $\rightarrow$ concatenate 4-way logits (8 dims) $\rightarrow$ meta-MLP ($8 \rightarrow 32 \rightarrow$ sigmoid)
 - b. Framewise
 - Majority-vote ensemble



Attention Based Methods

- Attention Fusion (Non-Framewise)
 - Project text/audio/video to 128d
 - Fuse via 2-layer, 4-head Transformer → mean-pool → 128 d
 - MLP → 1 → sigmoid
- Framewise Cross-Attention
 - Concat audio+video per frame → project to 512d
 - Bi-GRU encode context & utterance
 - Cross-attention (utterance queries context) → attention-pool → 512 d
 - Fuse with projected text → MLP → sigmoid



Results

Method	Accuracy	Precision	Recall	F1 Score
Early Fusion	0.7101	0.667	0.8571	0.7500
Late Fusion Ensemble	0.6812	0.6857	0.6857	0.6857
Attention Fusion (aggregated embeddings)	0.710	0.692	0.771	0.730
Early Fusion (Framewise)	0.6667	0.6579	0.7143	0.6849
Late Fusion Voting (Framewise)	0.594	0.569	0.829	0.674
Attention Fusion (Framewise)	.696	.667	.800	.727
Gemini 2.0 Flash (for comparison)	0.6812	0.6275	0.9143	0.7442
Gemini 2.0 Pro (for comparison)	0.7971	0.7143	1.00	0.833
MUSTARD SVM (baseline)	(Not in Paper)	0.643	0.626	0.628

- Comparison with Gemini results go to show how nuanced the problem really is
- GRUs didn't perform well->may require more training data for better sequence understanding, same for the attention-based methods
- Late Fusion- may not capture the nuances of modalities in great-depth (deep modeling would be the way to go)

Methods(OLD keeping for reference) Will delete slide later

- **Transformer-based Sentence Embeddings**
 - Experiment with various transformer architectures to produce sentence-level embeddings tailored to sarcasm detection
 - Use SBERT sentence_transformers library to facilitate the implementation of SOTA models
- **Audio** - Used the WavLM model for deep feature embedding, Librosa library to extract spectral features such as MFCC, Parselmouth to extract pitch and intensity features
- **Visual-**
 - BlazeFace x Attention Mesh (Google MediaPipe) for extracting 478 3D Facial landmarks/frame & stack together per video
 - DinoV2 (Self-Supervised ViT variant) and to extract embeddings/frame & stack together per video
- **Test context handling methods:** concatenating context and final clips with a separation token or treating them as a single clip
- **Align and synchronize-** audio, visual, and textual data for consistency.
- **Adapt methodology used in the FABa paper (ECCV 2024) on our video+audio+text data** [they use image data with text] for sarcasm detection using our multiple modalities *by fine-tuning a VLM.*

Summary

- Contributions split-
 - Audio and Attention Fusion: Abi
 - Visual and Early Fusion: Anupam
 - Text and Late Fusion: Nick
- Problem has great scope to be worked on-
 - Better training data
 - Using and improving VLMs for this task
 - Better representational methodologies for capturing Sarcasm
- Learnings-
 - Understanding of fusion & modality representational methods, modeling techniques
 - Better understanding of Sarcasm (?)

